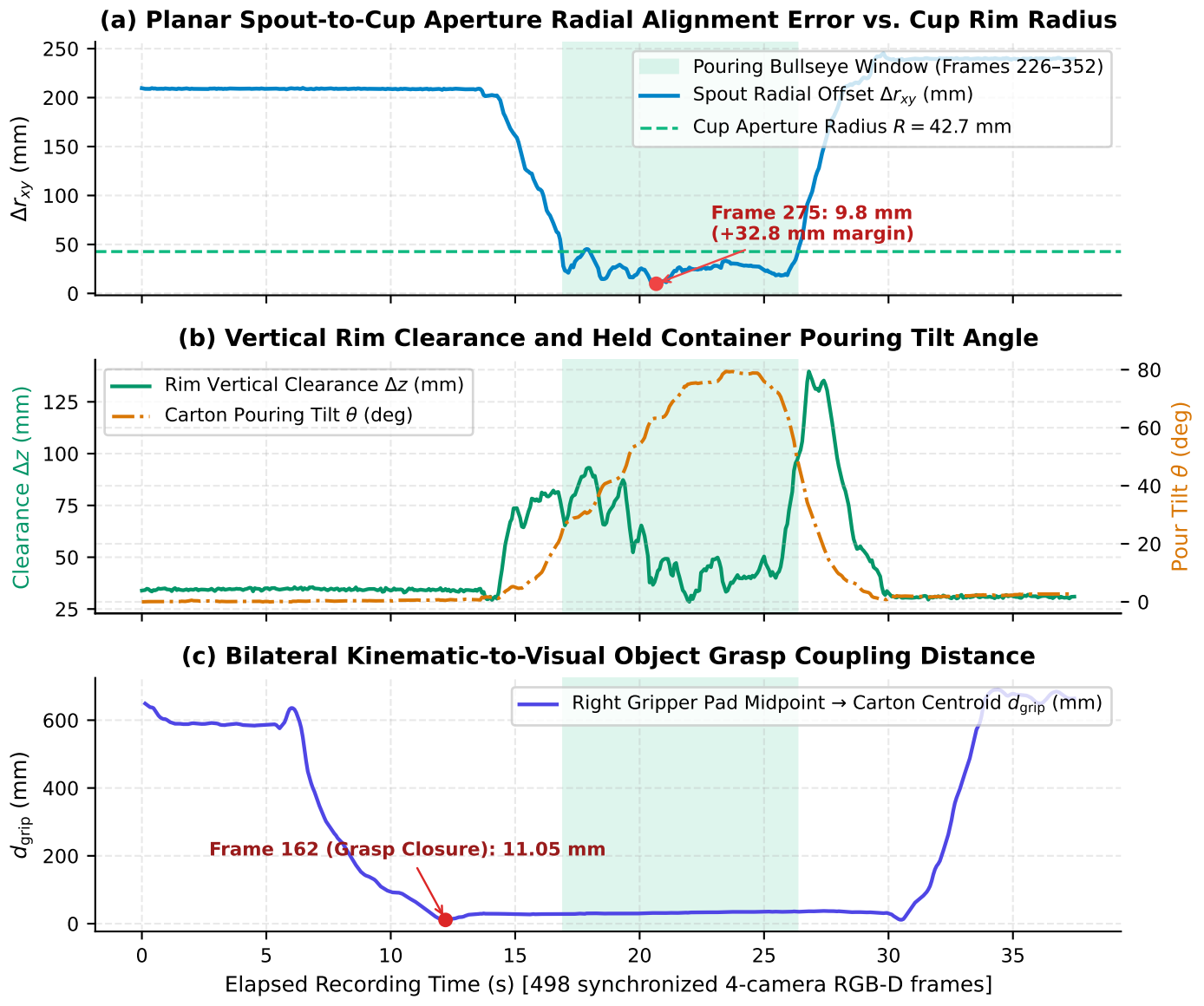
(a) Planar Spout-to-Cup Aperture Radial Alignment Error vs. Cup Rim Radius
Pouring Bullseye Window (Frames 226–352)
Spout Radial Offset Δr_{xy} (mm)
Cup Aperture Radius $R = 42.7$ mm
Frame 275: 9.8 mm
(+32.8 mm margin)
Δr_{xy} (mm)
250
200
150
100
50
0
(b) Vertical Rim Clearance and Held Container Pouring Tilt Angle
Rim Vertical Clearance Δz (mm)
Carton Pouring Tilt θ (deg)
Clearance Δz (mm)
Pour Tilt θ (deg)
125
100
75
50
25
80
60
40
20
0
(c) Bilateral Kinematic-to-Visual Object Grasp Coupling Distance
Right Gripper Pad Midpoint → Carton Centroid d_{grip} (mm)
Frame 162 (Grasp Closure): 11.05 mm
d_{grip} (mm)
600
400
200
0
0
5
10
15
20
25
30
35
Elapsed Recording Time (s) [498 synchronized 4-camera RGB-D frames]